



BITS Pilani

Automotive Vehicle

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Automotive Vehicle (AEL ZC441)

Lecture: Powertrain Fundamentals



Content of Automotive Vehicle?

Course Overview



- **Course Title**

Automotive Vehicles (AEL ZC441) - 3 Credit Units, 32-hour contact model.

- **Course Description**

Internal combustion engines; vehicle performance; analysis and design of vehicle components. Experimental or theoretical investigation of problems selected from the field of automotive vehicles.

- **Course Objectives**

Study of main automobile systems (engine, transmission, drive-axle, suspension, brakes); understanding working principles; performance analysis; and exposure to advanced concepts through projects and assignments.

- **Textbooks**

T1: S. Srinivasan, Automotive Mechanics - Tata McGraw-Hill. T2: Automotive Handbook, 11th Edition - Bosch GMBH, Wiley.



How We Will Proceed

- **Format**

16 contact sessions delivered over the semester, each built around one focused topic from the Learning Plan.

- **Self-study expectation**

Students are responsible for maintaining the self-study schedule alongside contact sessions, per the course handout.

- **eLearn Portal**

All announcements, deadlines, and session material are posted on the BITS eLearn portal - check it regularly.

- **Assessment rhythm**

Continuous evaluation through quizzes/assignments (EC1), a Mid-Semester Test (EC2) after Session 8, and a Comprehensive Exam (EC3) covering all topics at the end.

- **Given this cohort's experience**

Sessions will move quickly through foundational material and spend more time on industry practice, trade-offs, and current developments.

Evaluation Scheme



Component	Type	Weight
EC1 - Quiz	Online	10%
EC1 - Assignment	Online	20%
EC2 - Mid-Semester Test	Closed Book	30%
EC3 - Comprehensive Exam	Open Book	40%

16 Session Learning Plan (Sessions 1-8)

Session	Topic
1	Introduction to Automobiles - classification, homologation, development methodology, virtual engineering
2	Chassis and Body - types of chassis and applications, weight distribution and analysis
3	Powertrain Fundamentals - propulsion requirements, tractive force and torque, powertrain architecture
4, 5, 6	IC Engine Fundamentals - engine basics, fuel conversion, torque characteristics, emissions
7	Electric Drive & Alternate Fuel Vehicles - BEV vs HEV, alternate fuel, fuel cell/hydrogen vehicles
8	Transmission Systems - manual, epicyclic, CVT, dual-clutch, AMT gearboxes

16 Session Learning Plan (Sessions 9-16)

Session	Topic
9	Vehicle Control Fundamentals - E/E architecture, sensors and actuators, ECU architecture
10	Cooling & Lubrication Systems - heat generation in ICE and EV, lubrication
11	Vehicle Dynamics - fundamentals, influencing factors, numerical problems
12	Suspension - fundamentals, types, impact on dynamics
13	Steering Systems - fundamentals, dynamics, EPAS and SBW systems
14	Brakes - fundamentals, impact on dynamics, vacuum assist systems
15	Passive Safety - airbags, seatbelts, ABS systems
16	Active Safety Systems - ADAS and driver assist systems